

```

106.                 else:
107.                     pointsToDelete.append((startx, y))
108.
109.             elif orientation == 'horizontal':
110.                 # Go up one path from the start point, and
111.                 # see if we can remove this segment:
112.                 for changex in (-1, 1):
113.                     x = startx
114.                     while 0 < x < width - 1:
115.                         x += changex
116.                         if (canvas[(x, starty - 1)] == BLACK and
117.                             canvas[(x, starty + 1)] == BLACK):
118.                             # We've found a four-way intersection.
119.                             break
120.                         elif ((canvas[(x, starty - 1)] == WHITE and
121.                             canvas[(x, starty + 1)] == BLACK) or
122.                             (canvas[(x, starty - 1)] == BLACK and
123.                             canvas[(x, starty + 1)] == WHITE)):
124.                             # We've found a T-intersection; we can't
125.                             # delete this segment:
126.                             canDeleteSegment = False
127.                             break
128.                     else:
129.                         pointsToDelete.append((x, starty))
130.                 if not canDeleteSegment:
131.                     continue # Get a new random start point.
132.                 break # Move on to delete the segment.
133.
134.             # If we can delete this segment, set all the points to white:
135.             for x, y in pointsToDelete:
136.                 canvas[(x, y)] = WHITE
137.
138.         # Add the border lines:
139.         for x in range(width):
140.             canvas[(x, 0)] = BLACK # Top border.
141.             canvas[(x, height - 1)] = BLACK # Bottom border.
142.         for y in range(height):
143.             canvas[(0, y)] = BLACK # Left border.
144.             canvas[(width - 1, y)] = BLACK # Right border.
145.
146.         # Paint the rectangles:
147.         for i in range(numberOfRectanglesToPaint):
148.             while True:
149.                 startx = random.randint(1, width - 2)
150.                 starty = random.randint(1, height - 2)
151.
152.                 if canvas[(startx, starty)] != WHITE:
153.                     continue # Get a new random start point.
154.                 else:
155.                     break
156.
157.         # Flood fill algorithm:
158.         colorToPaint = random.choice([RED, YELLOW, BLUE, BLACK])
159.         pointsToPaint = set([(startx, starty)])
160.         while len(pointsToPaint) > 0:

```